

Carboxylic Acid and Derivatives

UNIT XIV

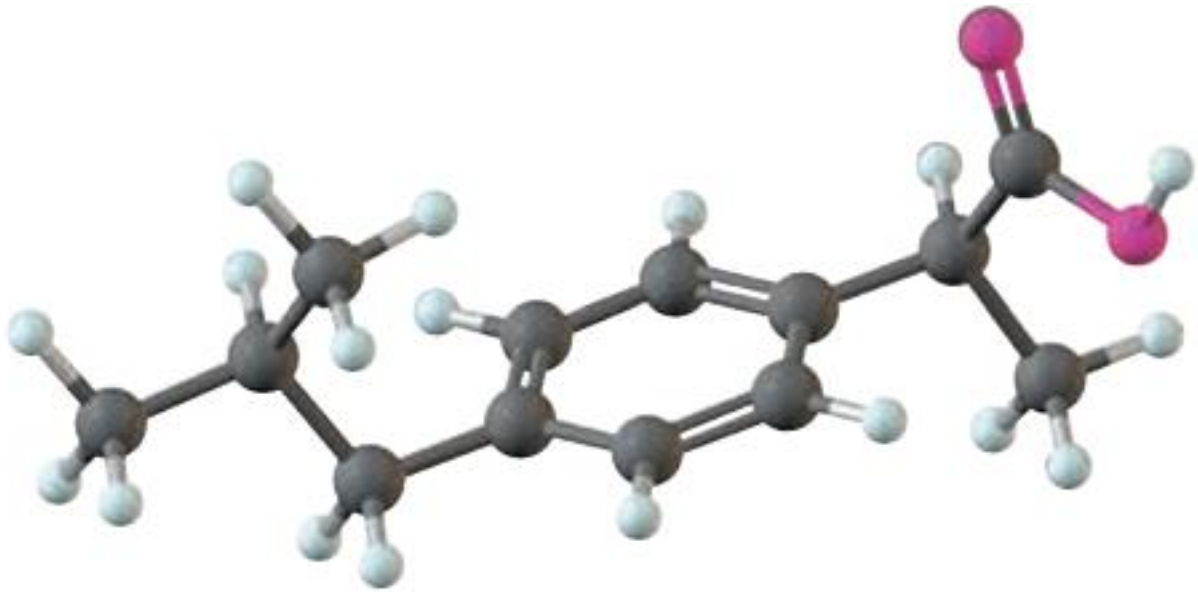
Learning Outcomes

At the end of each topic, the students should be able to:

- Identify and classify organic compounds of carboxylic acid and derivatives based on their structures and characteristic functional groups.
- Name and write the structural formulas of carboxylic acid and derivatives using appropriate IUPAC nomenclature rules.
- Explain the physical properties of carboxylic acid and derivatives .
- Describe the characteristic chemical reactions of carboxylic acid and derivatives based on their functional groups.

OVERVIEW





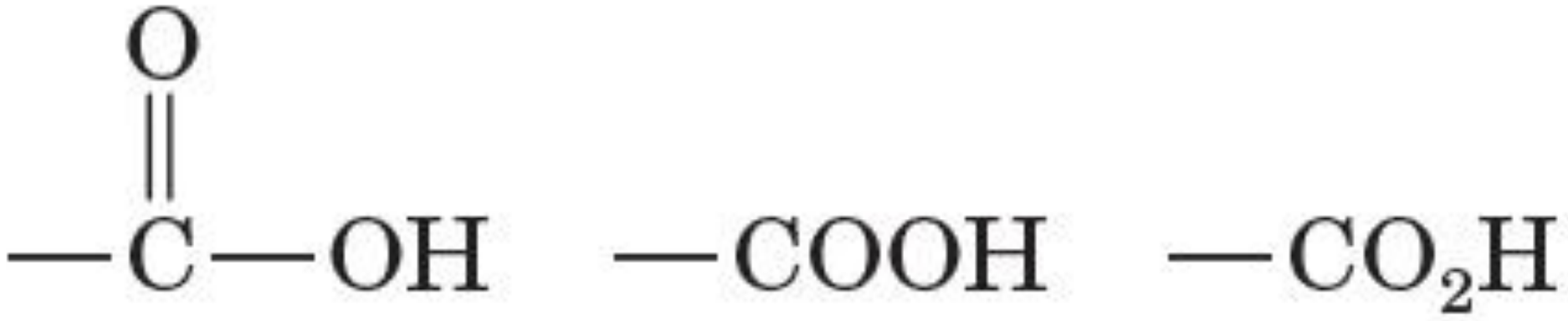
a model of (S)-ibuprofen

The active ingredients in these two nonprescription pain relievers are derivatives of arylpropanoic acids.

IMAGE SOURCE: Brown, W., Iverson, B., Anslyn, E., and Foote, C. (2023) Organic Chemistry 9th ed. Cengage Learning.

Carboxylic Acids

- Another class of organic compounds containing the carbonyl group.
- The functional group of a carboxylic acid is a **carboxyl group,** which can be represented in any one of three ways.
- It is made up of a carbonyl group and a hydroxyl group



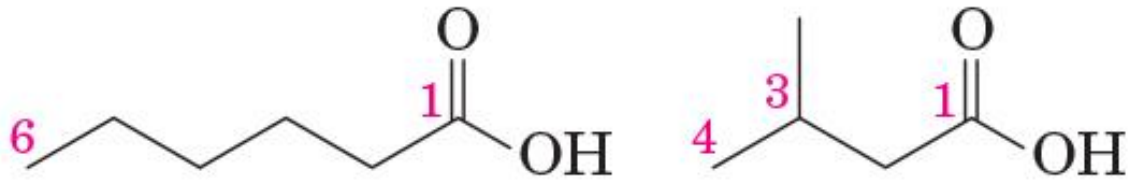
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Nomenclature

IUPAC names:

- For an acyclic carboxylic acid, take the longest carbon chain that contains the carboxyl group as the parent alkane.
- Drop the final **-e** from the name of the parent alkane and replace it by **-oic acid.**
- Number the chain beginning with the carbon of the carboxyl group.
- Because the carboxyl carbon is understood to be carbon 1, there is no need to give it a number.

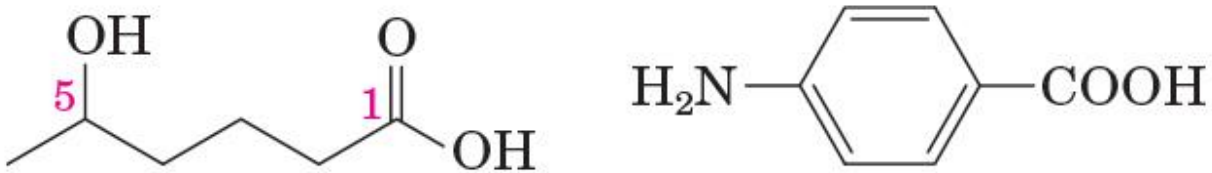
- In these examples, the common name is given in parentheses.



Hexanoic acid (Caproic acid)

3-Methylbutanoic acid (Isovaleric acid)

- An –OH substituent is indicated by the prefix **hydroxy-**; an $–NH_2$ substituent by the prefix **amino-.**

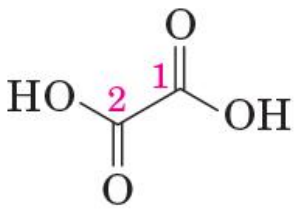


5-Hydroxyhexanoic acid

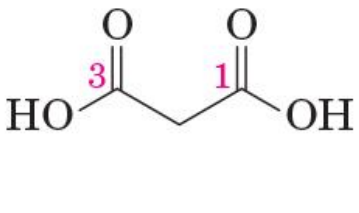
4-Aminobenzoic acid (*p*-Aminobenzoic acid)

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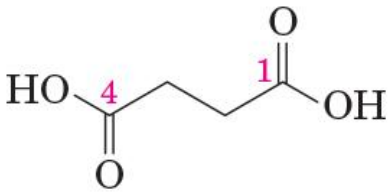
- To name a dicarboxylic acid, add the suffix -**dioic acid** to the name of the parent alkane that contains both carboxyl groups; thus, **-ane** becomes **-anedioic acid.**
- The numbers of the carboxyl carbons are not indicated because they can be only at the ends of the chain.



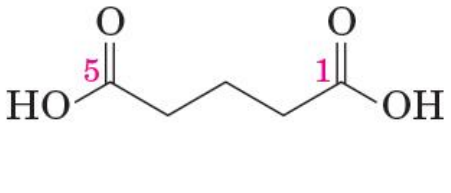
Ethanedioic acid
(Oxalic acid)



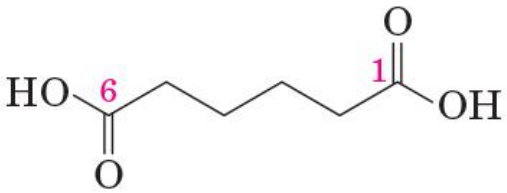
Propanedioic acid
(Malonic acid)



Butanedioic acid
(Succinic acid)



Pentanedioic acid
(Glutaric acid)



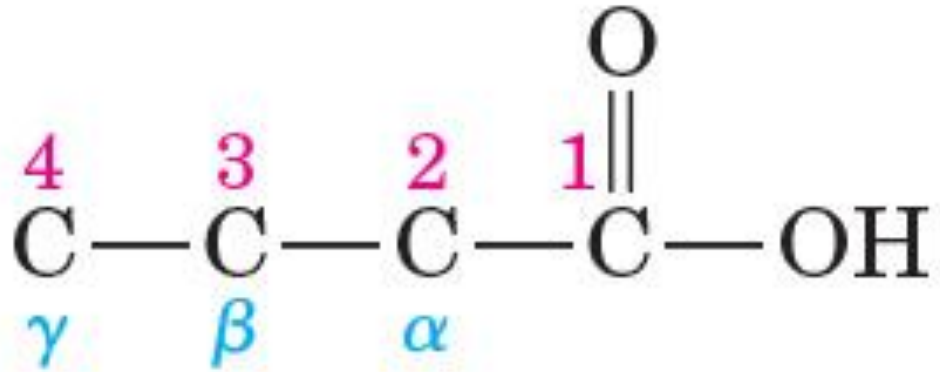
Hexanedioic acid
(Adipic acid)

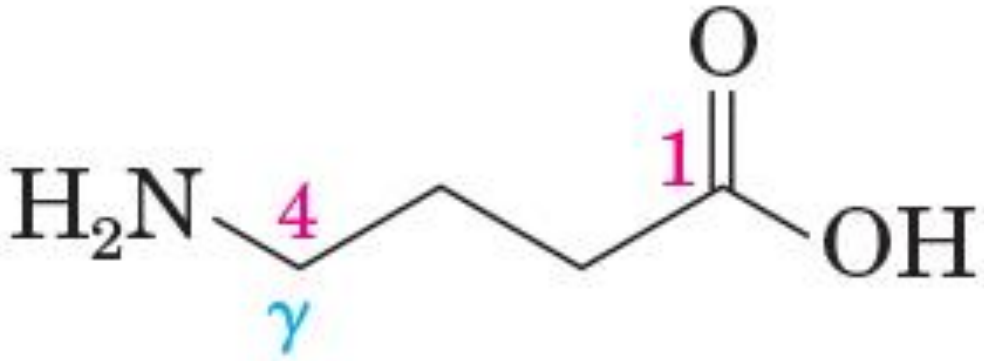
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Nomenclature: Table 17.1

Structure	IUPAC Name	Common Name	Derivation
HCOOH	methanoic acid	formic acid	Latin: *formica*, ant
CH_3COOH	ethanoic acid	acetic acid	Latin: *acetum,* vinegar
CH_3CH_2COOH	propanoic acid	propionic acid	Greek: *propion*, first fat
$CH_3(CH_2)_2COOH$	butanoic acid	butyric acid	Latin: *butyrum,* butter
$CH_3CH_2)_3COOH$	pentanoic acid	valeric acid	Latin: *valere,* to be strong
$CH_3(CH2)_4COOH$	hexanoic acid	caproic acid	Latin: *caper*, goat
$CH_3(CH_2)_6COOH$	octanoic acid	caprylic acid	Latin: *caper,* goat
$CH_3(CH_2)_8COOH$	decanoic acid	capric acid	Latin: *caper,* goat
$CH_3(CH_{2)10}COOH$	dodecanoic acid	lauric acid	Latin: *laurus,* laurel
$CH_3(CH_2)_{12}COOH$	tetradecanoic acid	myristic acid	Greek: *myristikos,* fragrant
$CH_3(CH_2)_{14}COOH$	hexadecanoic acid	palmitic acid	Latin: *palma*, palm tree
$CH_3(CH_2)_{16}COOH$	octadecanoic acid	stearic acid	Greek: *stear,* solid fat
$CH_3(CH_2)_{18}COOH$	eicosanoic acid	arachidic acid	Greek: *arachis*, peanut

- For common names, use, the Greek letters alpha (α), beta (β), gamma (γ), and so forth to locate substituents.





4-Aminobutanoic acid
(γ-Aminobutyric acid; GABA)

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EXERCISE

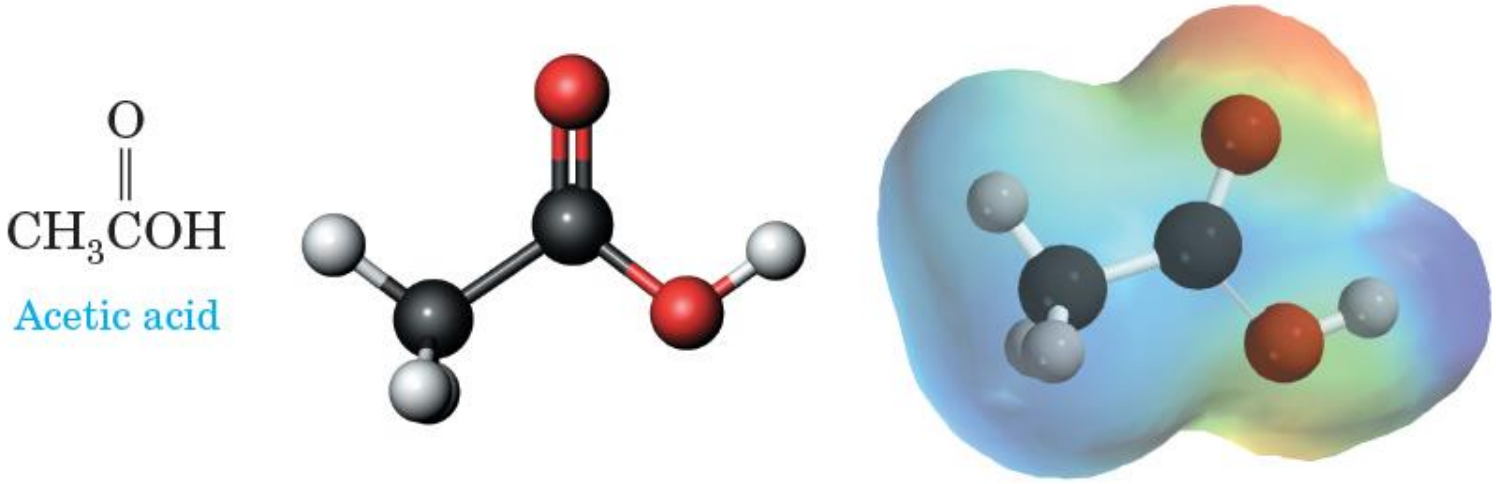
Megatomoic acid, the sex attractant of the female black carpet beetle, has the following structure. What is its IUPAC name?



3,5-tetradecadienoic acid

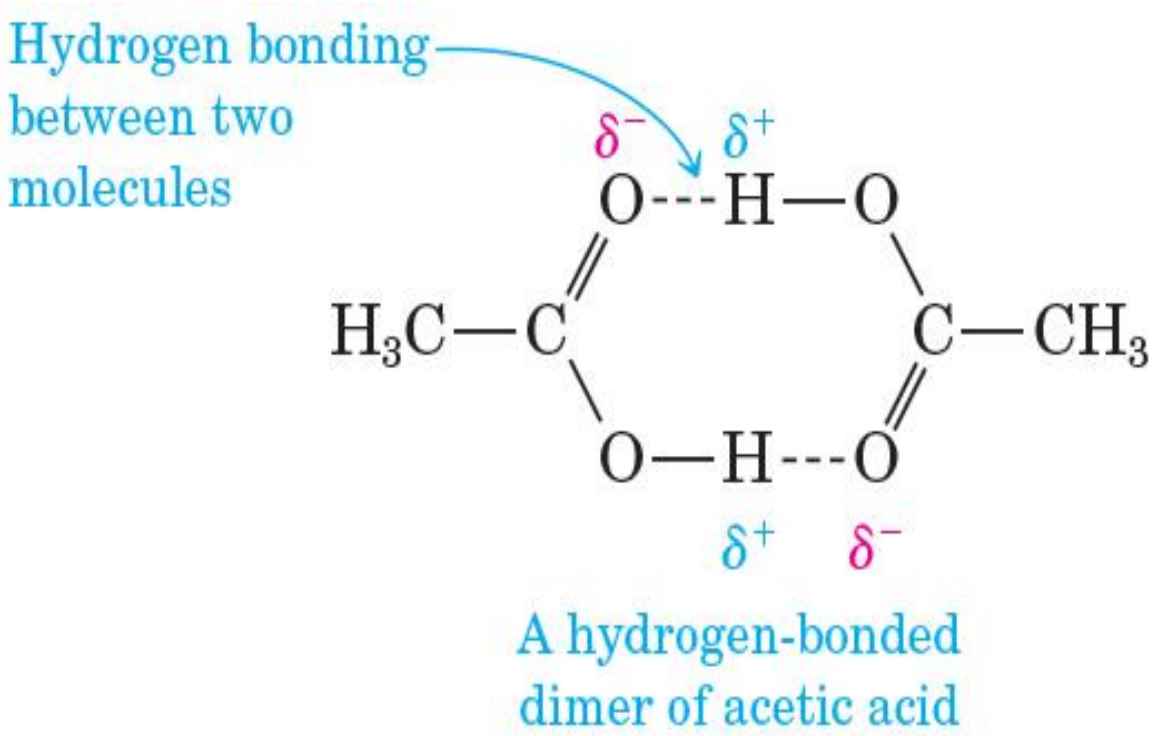
Physical Properties

- Figure 17.1 The carboxyl group contains three polar covalent bonds; C=O, C–O, and O–H.
- The polarity of these bonds determines the major physical properties of carboxylic acids.



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- Carboxylic acids have significantly higher boiling points than other types of organic compounds of comparable molecular weight.
- Their higher boiling points are a result of their polarity and the fact that hydrogen bonding between two carboxyl groups creates a dimer that behaves as a higher-molecular-weight compound.



A hydrogen-bonded dimer of acetic acid

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Table 17.2 Carboxylic acids are more soluble in water than are alcohols, ethers, aldehydes, and ketones of comparable molecular weight.

Structure	Name	Molecular Weight (amu)	Boiling Point (°C)	Solubility (g/100 mL H_2O)
CH_3COOH	acetic acid	60.1	118	Infinite
$CH_3CH_2CH_2OH$	1-propanol	60.1	97	Infinite
CH_3CH_2CHO	Propanal	58.1	48	16
$CH_3(CH_2)_2COOH$	butanoic acid	88.1	163	Infinite
$CH_3(CH_{2)3}CH_2OH$	1-pentanol	88.1	137	2.3
$CH_3(CH_2)_3CHO$	Pentanal	88.1	103	slight

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EXERCISE

Arrange the compounds in each set in the order of increasing boiling point.



(a)

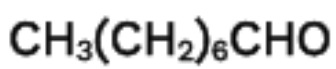


(b)

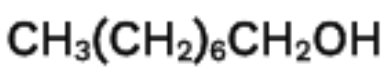


(c)

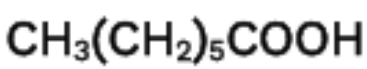
The following are listed in order of increasing boiling point:



bp 171°C



bp 195°C



bp 223°C

Fatty Acids

Fatty acids: Long chain carboxylic acids derived from animal fats, vegetable oils, or phospholipids of biological membranes.

- More than 500 have been isolated from various cells and tissues.
- Most have between 12 and 20 carbons in an unbranched chain.
- In most unsaturated fatty acids, the *cis* isomer predominates; *trans* isomers are rare.

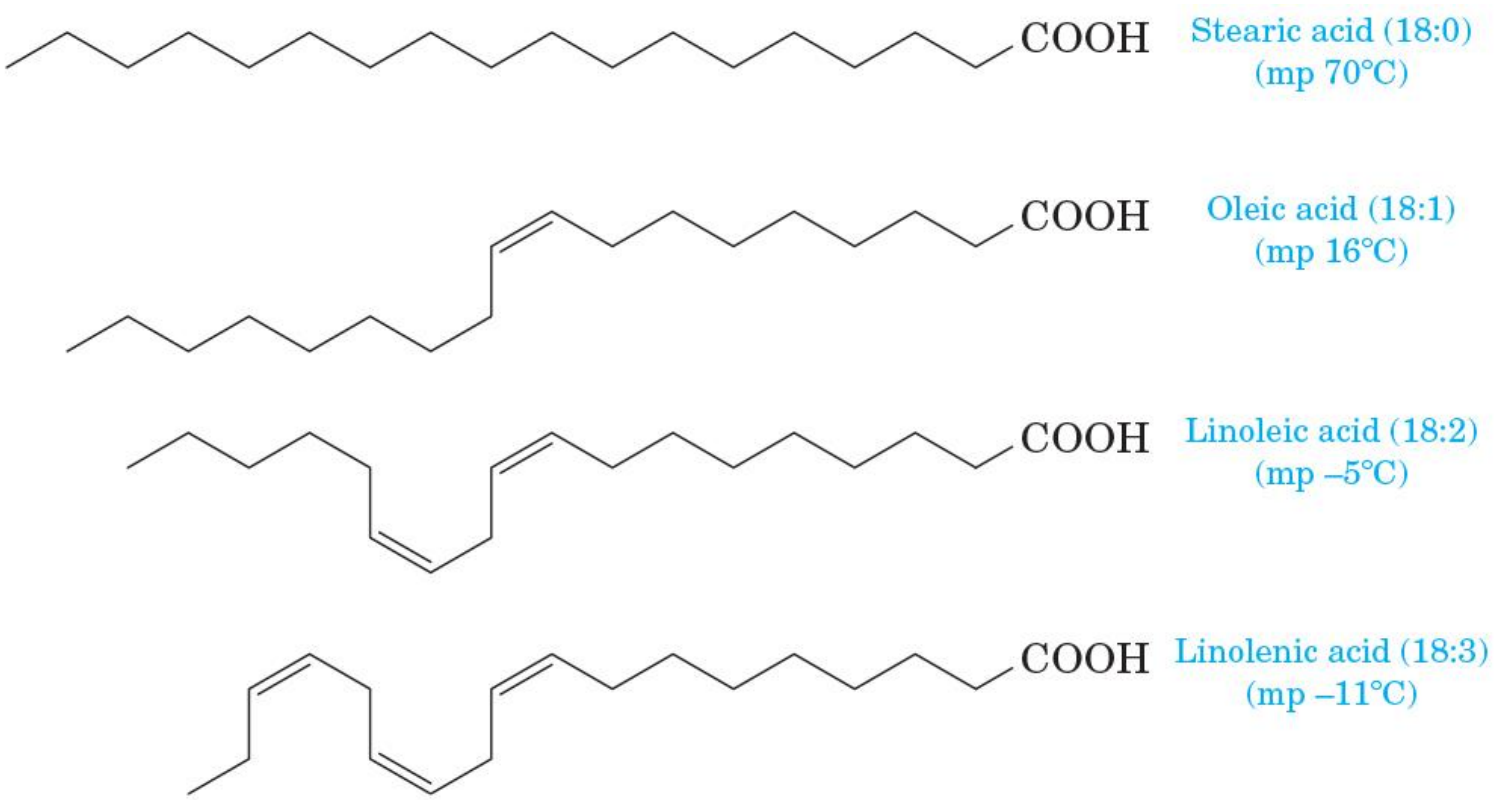
Table 17.3 The Most Abundant Fatty Acids in Animal Fats, Vegetable Oils, and Biological Membranes

Carbon Atoms: Double Bonds*	Structure	Common Name	Melting Point (°C)
Saturated Fatty Acids			
12:0	$CH_3(CH_2)_{10}COOH$	lauric acid	44
14:0	$CH_3(CH_2)_{12}COOH$	myristic acid	58
16:0	$CH_3(CH_2)_{14}COOH$	palmitic acid	63
18:0	$CH_3(CH_2)_{16}COOH$	stearic acid	70
20:0	$CH_3(CH_2)_{18}COOH$	arachidic acid	77
Unsaturated Fatty Acids			
16:1	$CH_3(CH_2)_5CH{=}CH(CH_2)_7COOH$	palmitoleic acid	1
18:1	$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$	oleic acid	16
18:2	$CH_3(CH_2)_4(CH{=}CHCH_2)_2(CH_2)_6COOH$	linoleic acid	-5
18:3	$CH_3CH_2(CH{=}CHCH_2)_3(CH_2)_6COOH$	linolenic acid	-11
20:4	$CH_3(CH_2)_4(CH{=}CHCH_2)_4(CH_2)_2COOH$	arachidonic acid	-49

* The first number is the number of carbons in the fatty acid; the second number is the number of carbon–carbon double bonds in its hydrocarbon chain.

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- **Unsaturated fatty acids** generally have lower melting points than their saturated counterparts.



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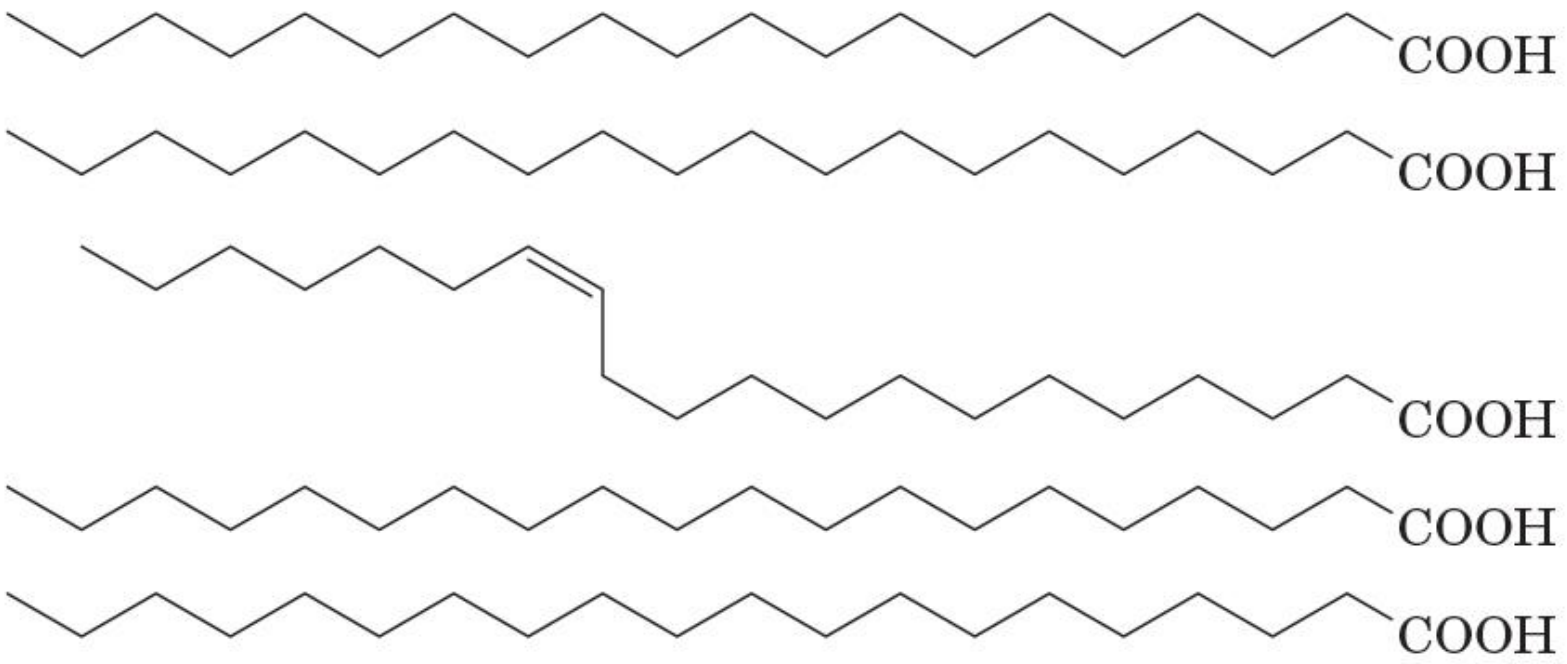
Saturated fatty acids are solids at room temperature.

- The regular nature of their hydrocarbon chains allows them to pack together in such a way as to maximize interactions (by London dispersion forces) between their chains.



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- In contrast, all unsaturated fatty acids are liquids at room temperature because the *cis* double bonds interrupt the regular packing of their hydrocarbon chains.

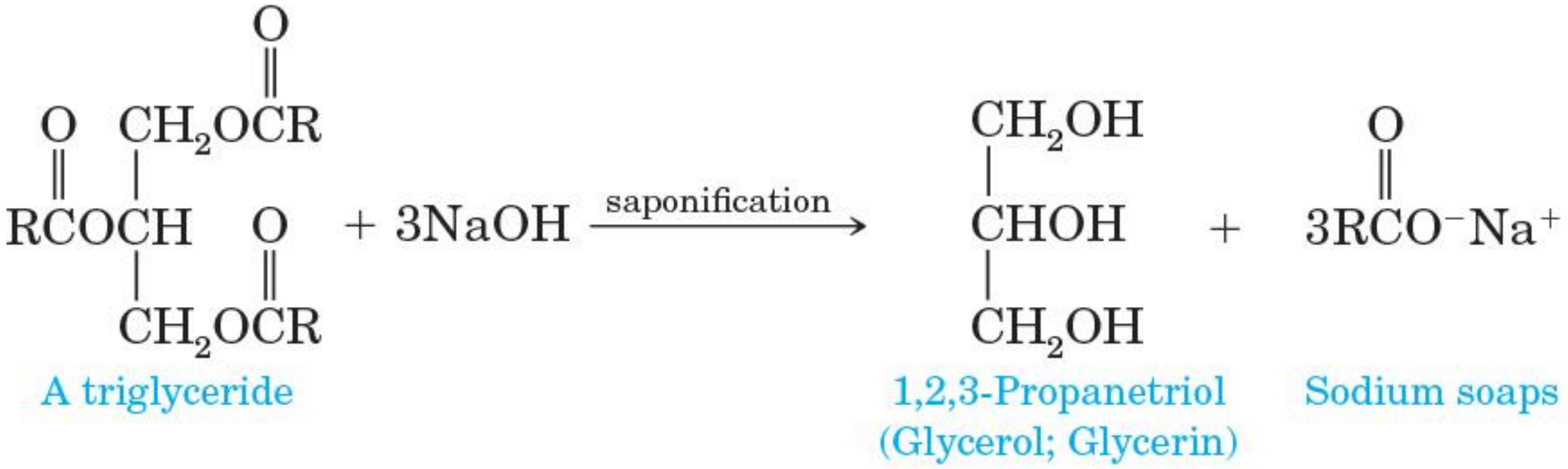


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Soaps

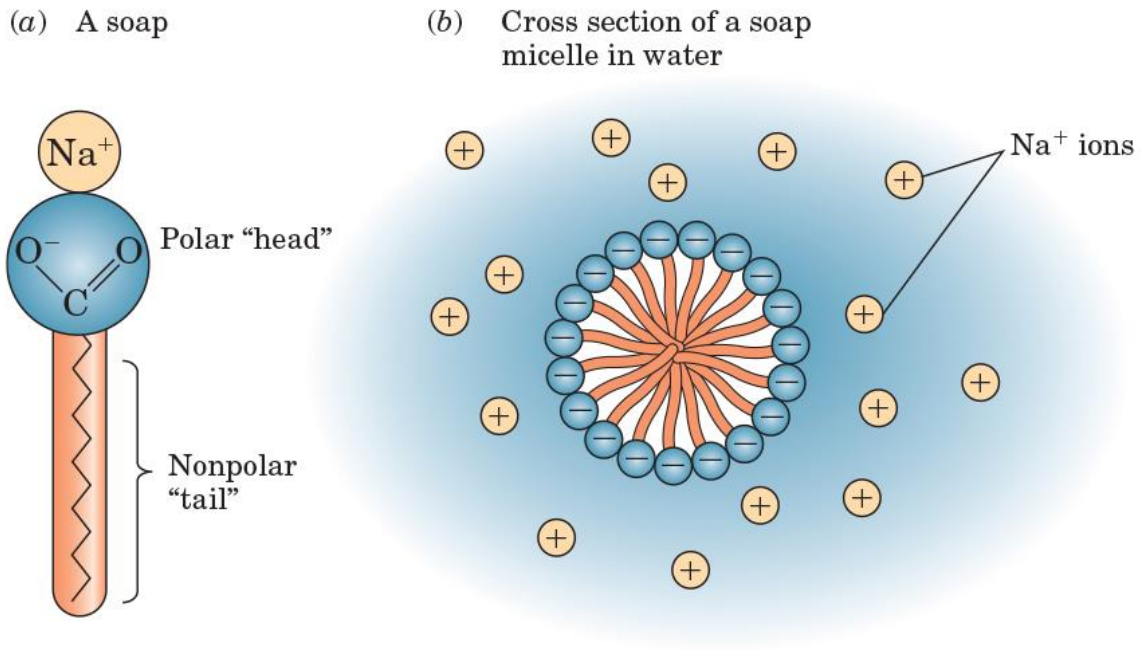
- Natural soaps are sodium or potassium salts of fatty acids.
- They are prepared from a blend of tallow and palm oils (triglycerides).
- Tallow is the insoluble triglyceride layer that forms on the top when solid fats from cattle are melted with steam.
- Triglycerides are triesters of glycerol.

- Preparation of soaps begins by boiling the triglycerides with NaOH. The reaction that takes place is called **saponification** (Latin: *saponem,* "soap"). Boiling with KOH gives a potassium soap.



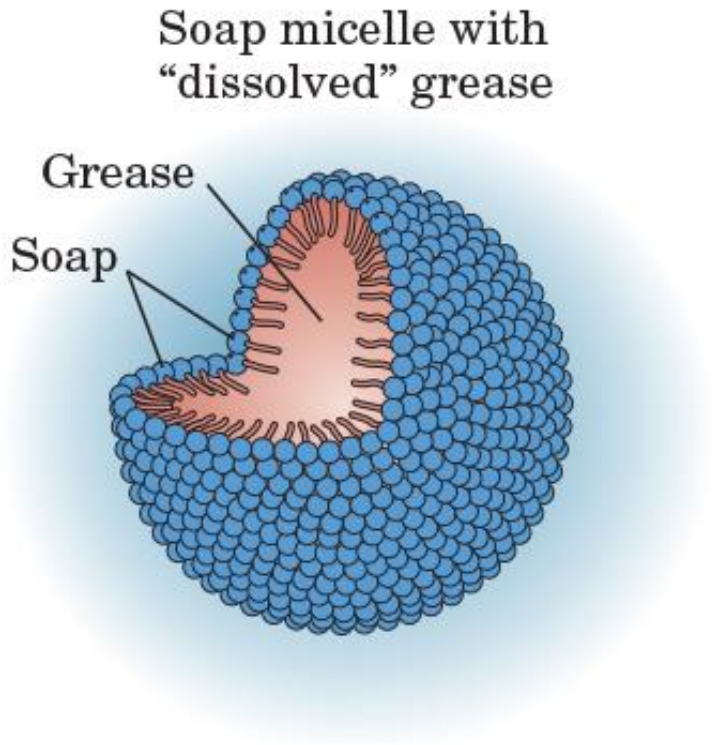
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Figure 17.2 In water, soap molecules spontaneously cluster into **micelles,** a spherical arrangement of molecules such that their **hydrophobic** parts are shielded from the aqueous environment, and their **hydrophilic** parts are in contact with the aqueous environment.



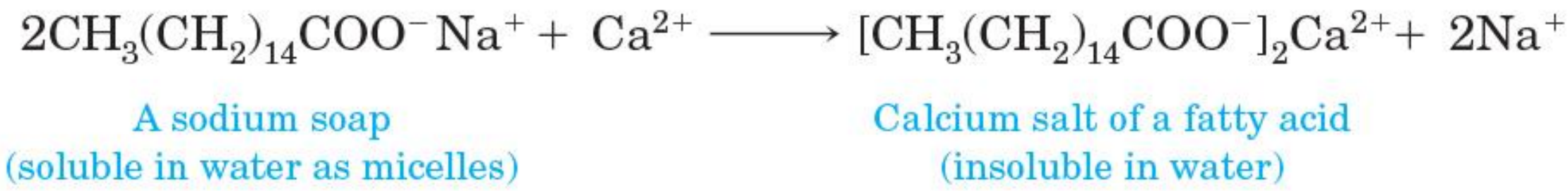
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Figure 17.3 When soaps and dirt, such as grease, oil, and fat stains are mixed in water, the nonpolar hydrocarbon inner parts of the soap micelles “dissolve” the nonpolar substances.



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- Natural soaps form water-insoluble salts in hard water.
- **Hard water** contains Ca^{2+}, Mg^{2+}, and Fe^{3+} ions.

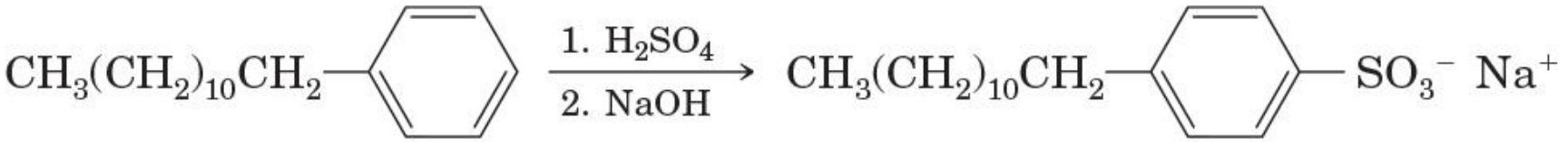


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Detergents

The problem of formation of precipitates in hard water was overcome by using a molecule containing a sulfonate ($-SO_3^-$) group in the place of a carboxylate ($-CO_2^-$) group.

- Calcium, magnesium and iron salts of sulfonic acids, RSO_3H, are more soluble in water than are their salts of fatty acids.
- Following is the preparation of the synthetic detergent, SDS, a linear alkylbenzenesulfonate (LAS), an anionic detergent.

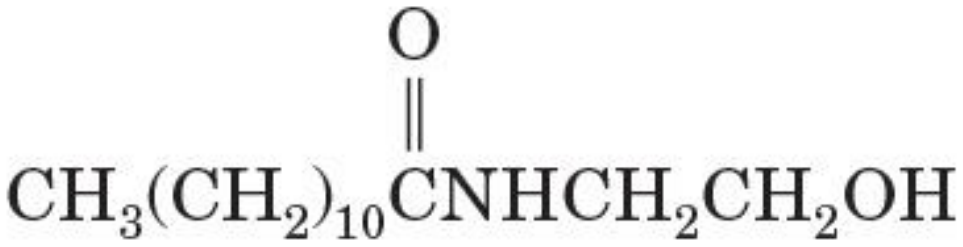


Dodecylbenzene

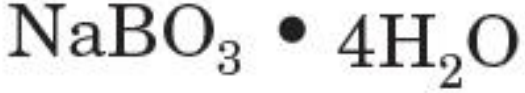
Sodium 4-dodecylbenzenesulfonate
(an anionic detergent)

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- Among the most common additives to detergents are foam stabilizers, bleaches, and optical brighteners.



N-(2-Hydroxyethyl)dodecanamide
(a foam stabilizer)



Sodium perborate tetrahydrate
(a bleach)

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EXERCISE

You are given a sample of solid cooking grease, which consists of three long chain carboxylic acids bound to glycerol through ester linkages. To this sample of this material, you add 95% ethanol and 8 M sodium hydroxide and heat it continuously for 10 minutes. What process did you perform and what are the major products?

a. Esterification; glycerol and three molecules of carboxylic acid salts classified as soaps

b. Esterification; glycerol and three molecules of carboxylic acids

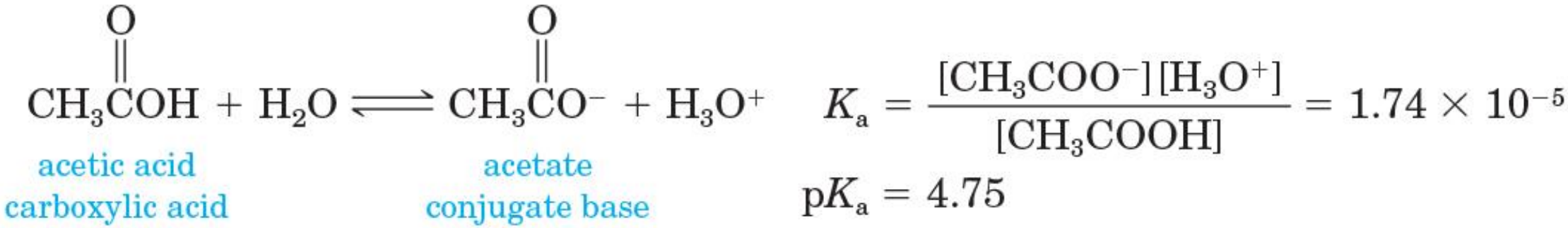
c. Saponification; glycerol and three molecules of carboxylic acid salts classified as soaps

d. Esterification; glycerol and three molecules of carboxylic acids

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Acidity of Carboxylic Acids

- Carboxylic acids are weak acids.
  - Values of K_a for most unsubstituted aliphatic and aromatic carboxylic acids fall within the range 10^{-4} to 10^{-5} (pK_a 4.0 – 5.0).



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Substituents of high electronegativity, especially –OH, –Cl, and $–NH_3^+$, near the carboxyl group increase the acidity of carboxylic acids.

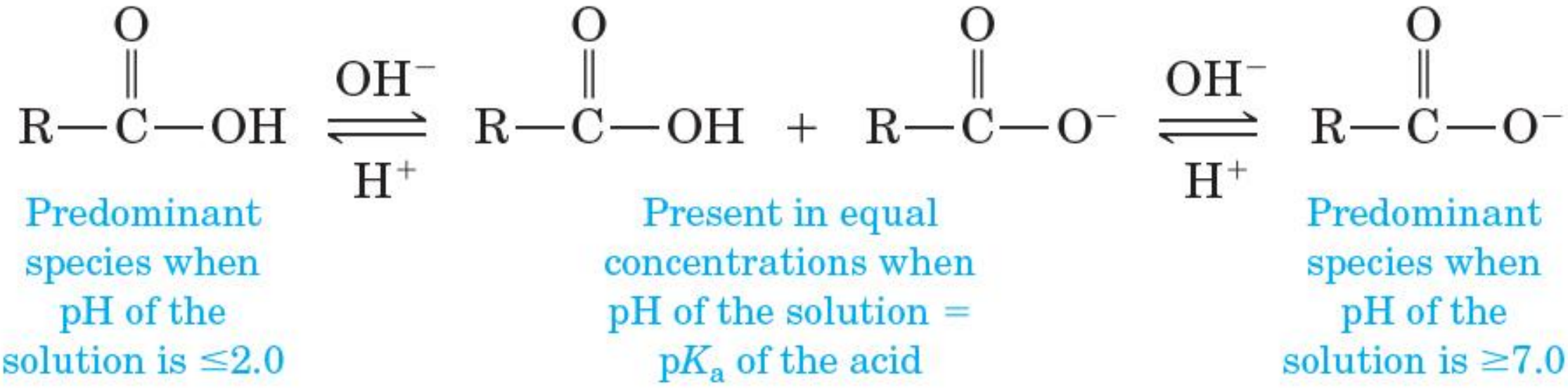
- Both dichloroacetic acid and trichloroacetic acid are stronger acids than H_3PO_4 (pK_a 2.1).

Formula:	CH_3COOH	$ClCH_2COOH$	$Cl_2CHCOOH$	Cl_3CCOOH
Name:	Acetic acid	Chloroacetic acid	Dichloroacetic acid	Trichloroacetic acid
pK_a:	4.75	2.86	1.48	0.70

Increasing acid strength

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- When a carboxylic acid is dissolved in aqueous solution, the form of the carboxylic acid present depends on the pH of the solution in which it is dissolved.

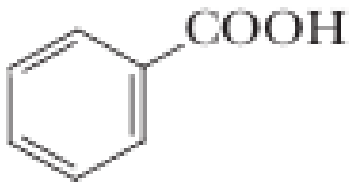


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EXERCISE

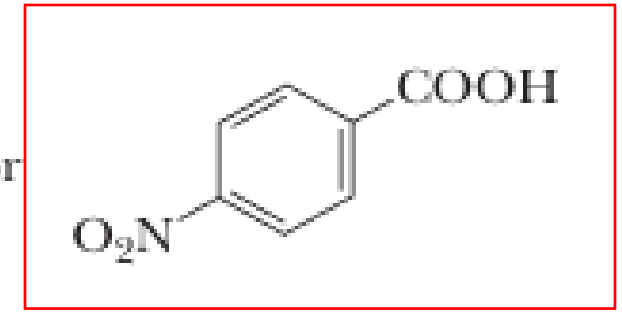
Which is the stronger acid in each pair?

(a)



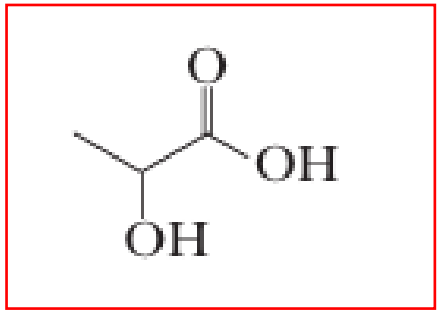
Benzoic acid

or



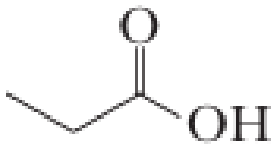
4-Nitrobenzoic acid

(b)



2-Hydroxypropanoic acid (Lactic acid)

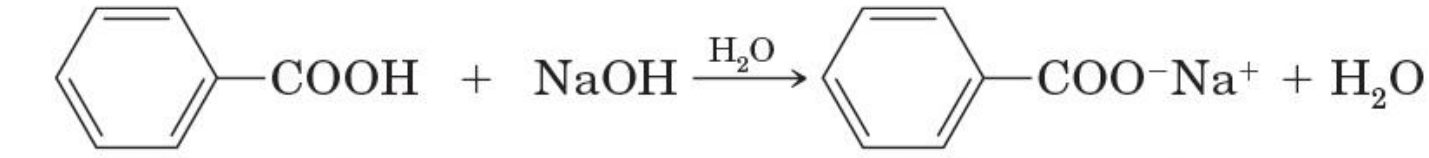
or



Propanoic acid

Reaction with Bases

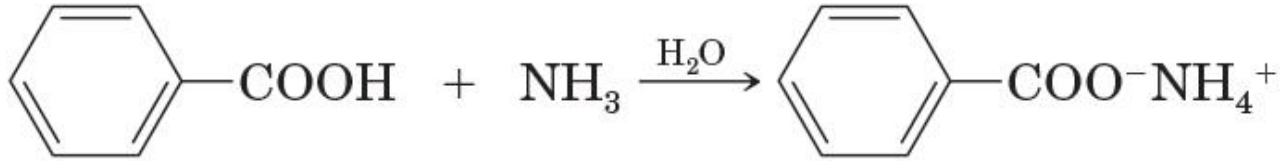
- All carboxylic acids, whether soluble or insoluble in water, react with NaOH, KOH, and other strong bases to form water-soluble salts.



Benzoic acid
(slightly soluble in water)

Sodium benzoate
(60 g/100 mL water)

- They also form water-soluble salts with ammonia and amines.



Benzoic acid
(slightly soluble in water)

Ammonium benzoate
(20 g/100 mL water)

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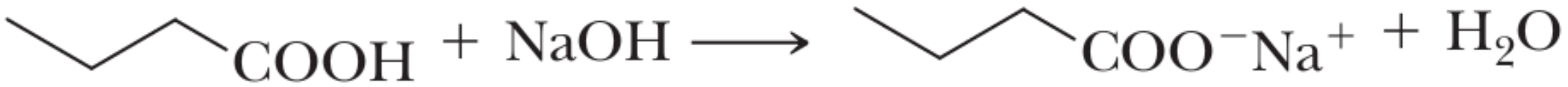
- Like inorganic acids, carboxylic acids react with sodium bicarbonate and sodium carbonate to form water-soluble sodium salts and carbonic acid.
- Carbonic acid then decomposes to give water and carbon dioxide, which evolves as a gas.

$$CH_3COOH(aq) + NaHCO_3(aq) \longrightarrow CH_3COO^-Na^+(aq) + CO_2(g) + H_2O(l)$$

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EXERCISE

Complete each acid-base reaction and name the carboxylic salt formed.

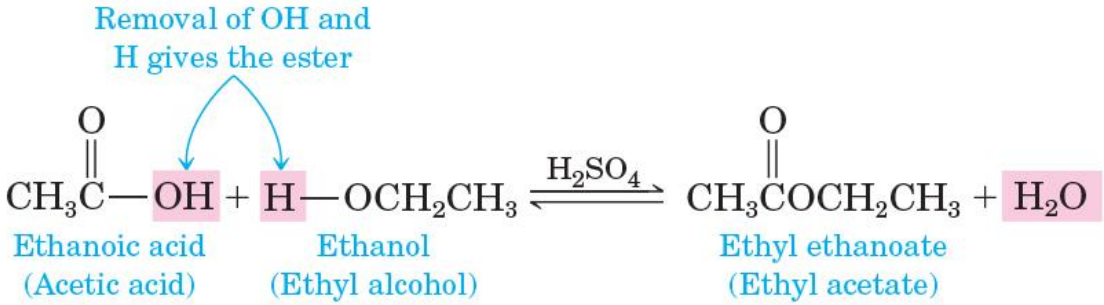


Butanoic acid **Sodium butanoate**

Fischer Esterification

Fischer esterification is one of the most commonly used methods for the preparation of esters.

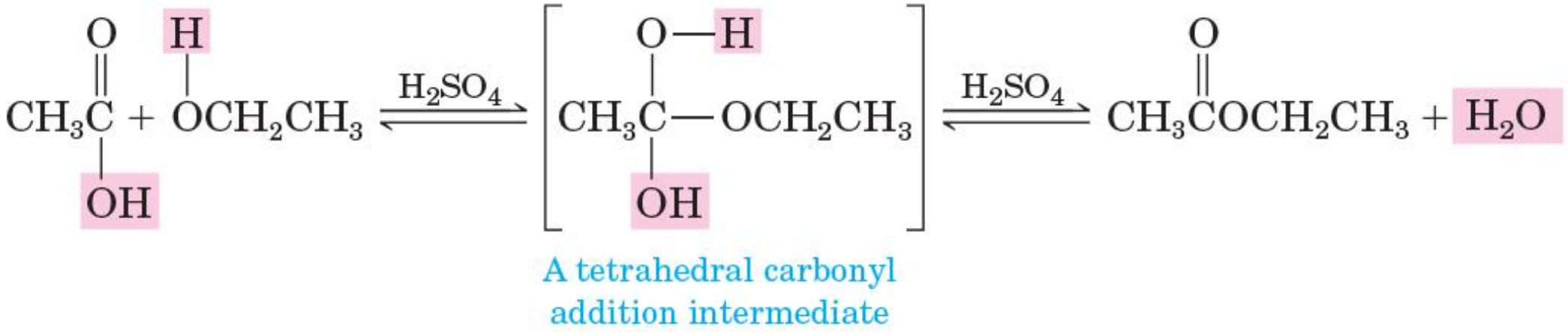
- In Fischer esterification, a carboxylic acid is treated with an alcohol in the presence of an acid catalyst, most commonly concentrated sulfuric acid.



  - Fischer esterification is reversible.
  - It is possible to drive it in either direction by the choice of experimental conditions (Le Chatelier's principle).

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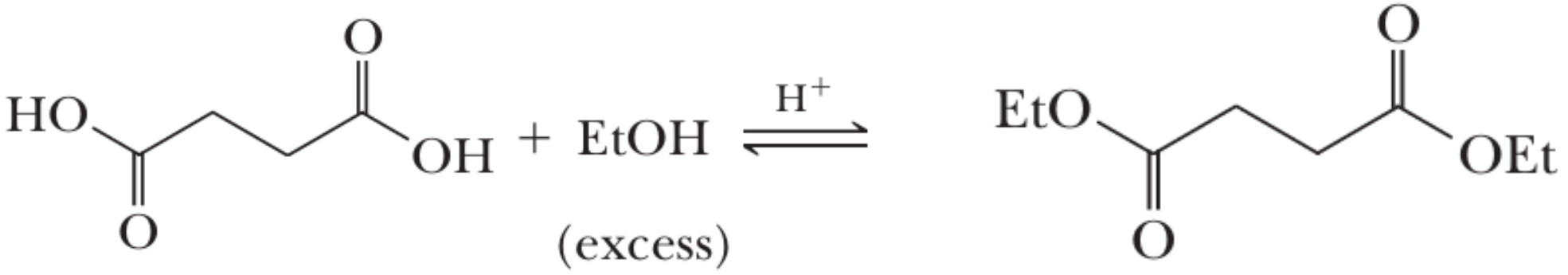
- In Fischer esterification, the alcohol adds to the carbonyl group of the carboxylic acid to form a tetrahedral carbonyl addition intermediate.
- The intermediate then loses H_2O to give an ester.



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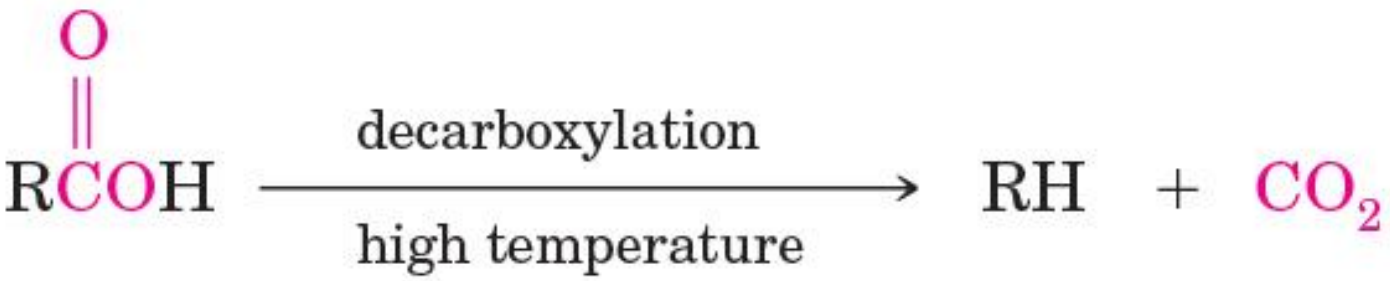
EXERCISE

Complete the equation for Fischer esterification reaction.



Decarboxylation

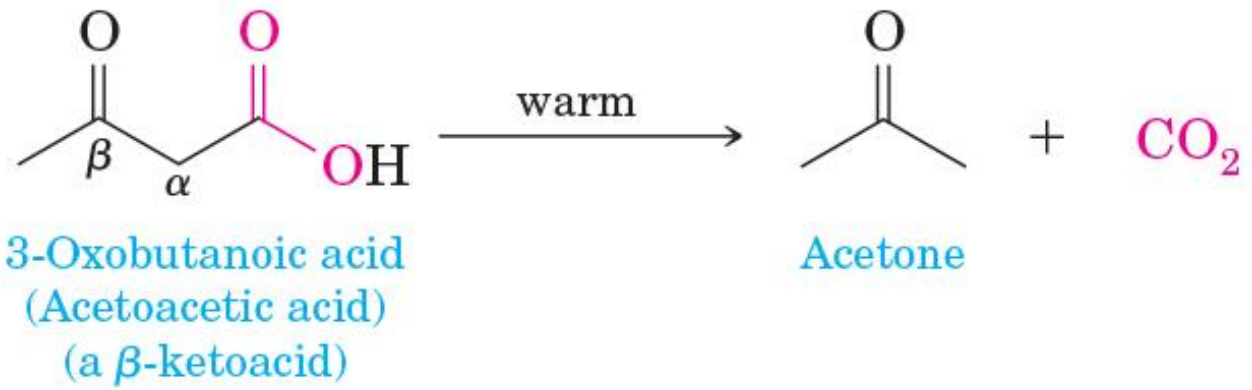
- **Decarboxylation:** The loss of CO_2 from a carboxyl group.
- Almost all carboxylic acids, when heated to a very high temperature, will undergo thermal decarboxylation.



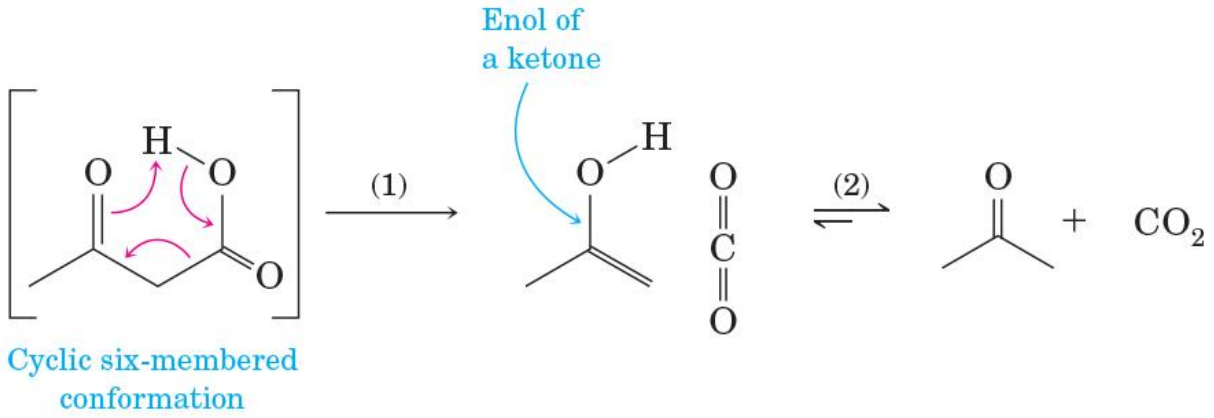
  - Most carboxylic acids, however, are resistant to moderate heat and melt and even boil without undergoing decarboxylation.
  - An exception is any carboxylic acid that has a carbonyl group on the carbon β to the COOH group.

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- Decarboxylation of a β-ketoacid.

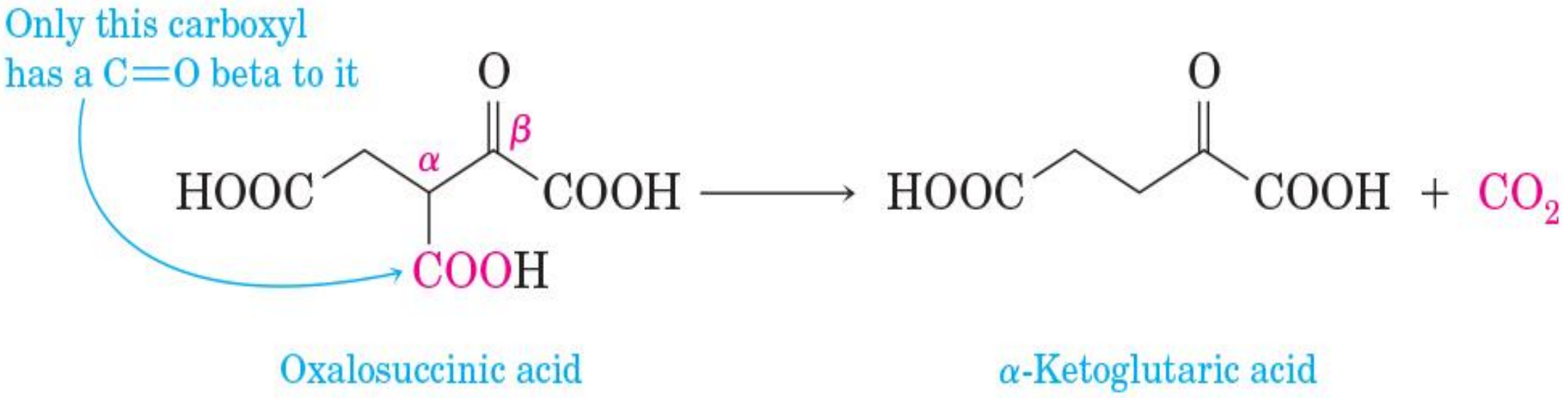


- The mechanism of thermal decarboxylation involves (1) redistribution of electrons in a cyclic transition state followed by (2) keto-enol tautomerism.



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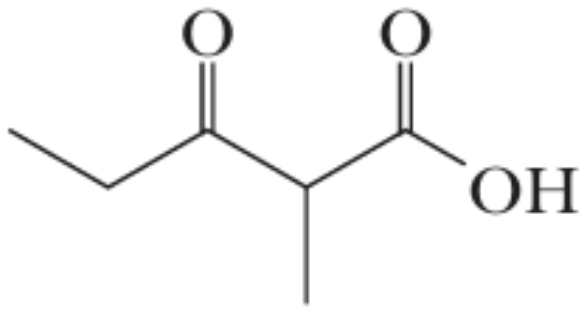
- An important example of decarboxylation of a β-ketoacid in biochemistry occurs during the oxidation of foodstuffs in the tricarboxylic acid (TCA) cycle. Oxalosuccinic acid, one of the intermediates in this cycle, has a carbonyl group (in this case a ketone) β to one of its three carboxyl groups.

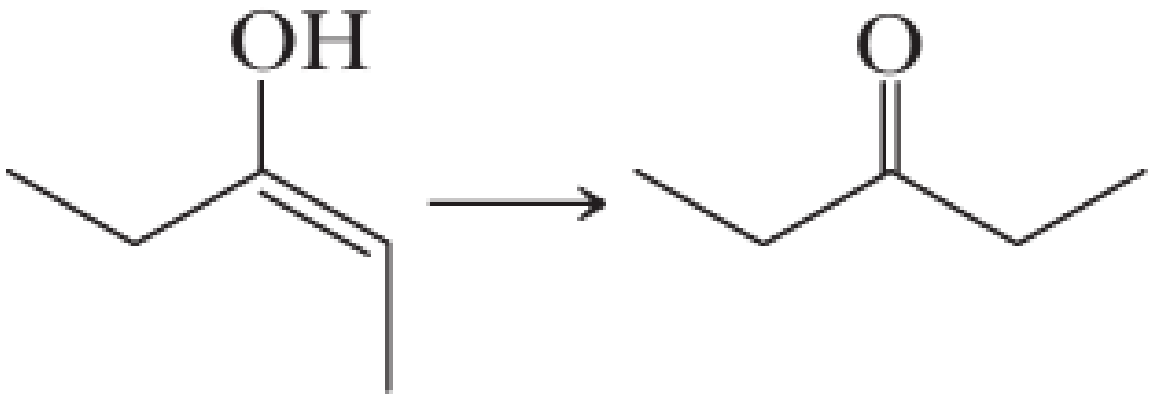


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EXERCISE

Each of these carboxylic acids undergoes thermal decarboxylation. Draw a structural formula for the enol intermediate and final product formed in each reaction.

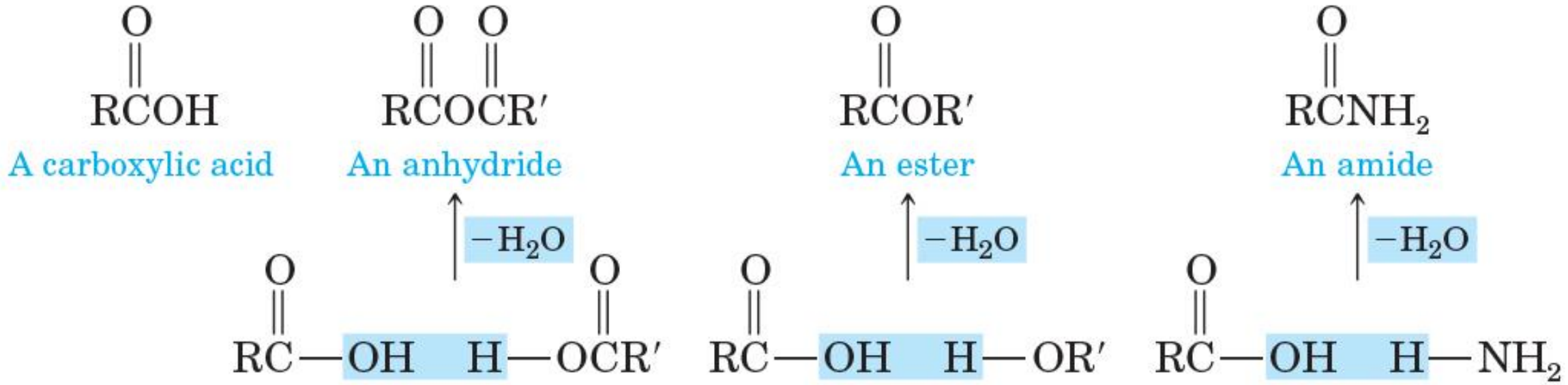




Carboxyl Derivatives

Three classes of compounds derived from carboxylic acids: **anhydrides, esters, and amides**.

- Each is related to a carboxyl group by loss of H_2O.

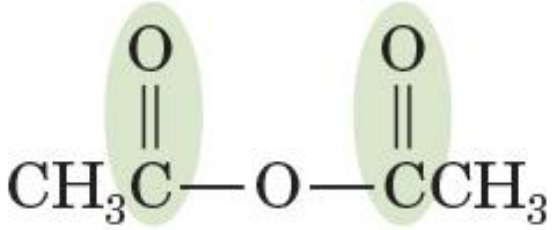


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Anhydrides

The functional group of an **anhydride** is two carbonyl groups bonded to the same oxygen.

- The anhydride may be symmetrical (from two identical acyl groups), or mixed (from two different acyl groups).
- To name an anhydride, drop the word "**acid**" from the name of the carboxylic acid from which the anhydride is derived and add the word "**anhydride**."



Acetic anhydride

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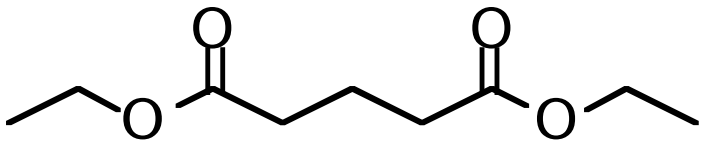
Esters

The functional group of an **ester** is a carbonyl group bonded to an –OR group. R may be alkyl or aryl.

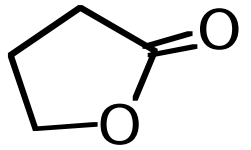
- Both IUPAC and common names of esters are derived from the names of the parent carboxylic acids.
- Name the alkyl or aryl group bonded to oxygen first, followed by the name of the acid; replace the suffix **-ic acid** by **-ate**.
- A cyclic ester is called a **lactone**.



Ethyl ethanoate (Ethyl acetate)



Diethyl pentanedioate (Diethyl glutarate)



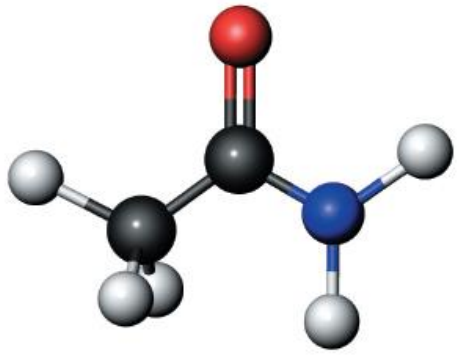
A five-membered lactone

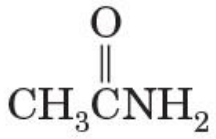
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Amides

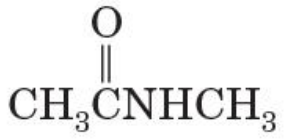
The functional group of an **amide** is a carbonyl group bonded to a nitrogen atom.

- To name an amide, drop the suffix **-oic acid** from the IUPAC name of the parent acid, or **-ic acid** from its common name, and add -amide.
- If the amide nitrogen is also bonded to an alkyl or aryl group, name the group and show its location on nitrogen by *N*-; two alkyl or aryl groups by *N,N*-di-.





Acetamide
(a 1° amide)



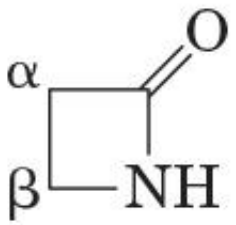
N-Methylacetamide
(a 2° amide)



N,N-Dimethylformamide
(a 3° amide)

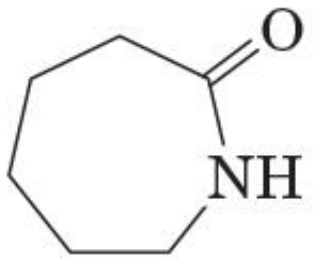
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A cyclic amide is called a **lactam**.



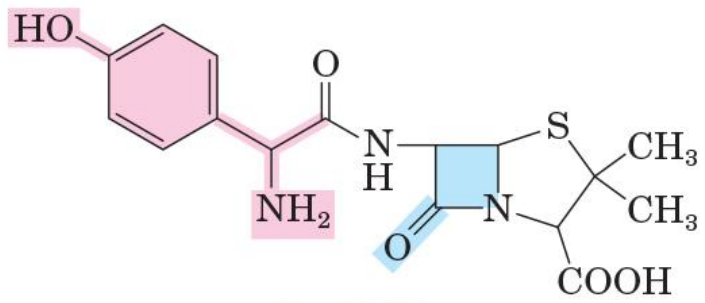
A four-membered lactam

(α β-lactam)



A seven-membered lactam

The penicillins are referred to as **β-lactam antibiotics**.

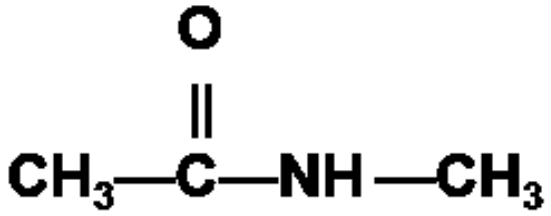


Amoxicillin

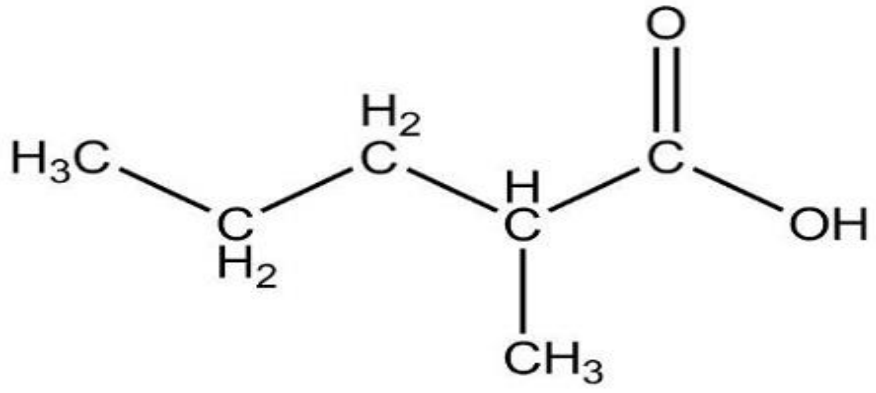
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EXERCISE

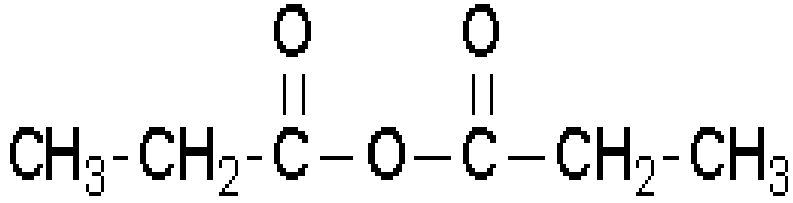
Write the IUPAC name of each organic compound.

1. 

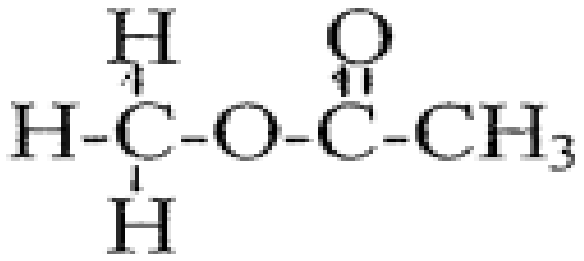
N-methylethanamide

2. 

2-methylpentanoic acid

3. 

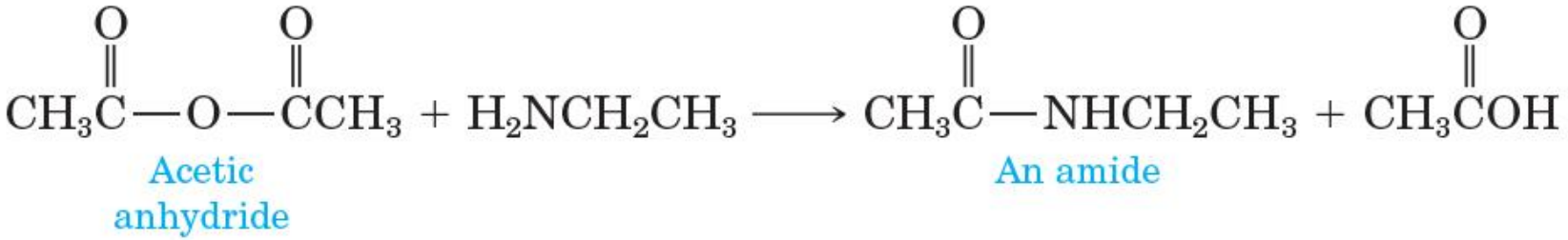
Propanoic anhydride

4. 

Methyl ethanoate

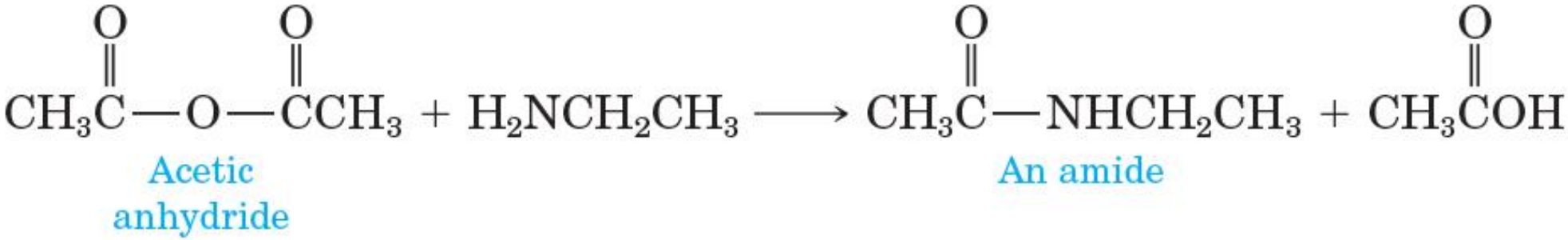
Preparation of Amides

- In principle, we can form an amide by treating a carboxylic acid with an amine and removing –OH from the acid and an –H from the amine.
  - In practice what occurs if the two are mixed is an acid-base reaction to form an ammonium salt.
  - If this salt is heated to a high enough temperature, water is eliminated and an amide forms.



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- It is much more common, however, to prepare amides by treating an anhydride with an amine.

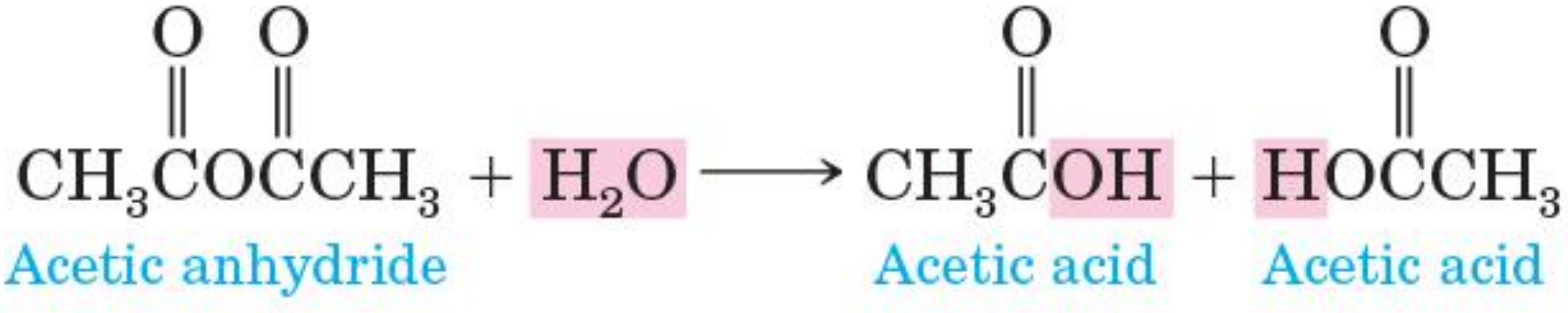


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Hydrolysis of Anhydrides

Hydrolysis is a chemical decomposition involving breaking a bond and the addition of the elements of water.

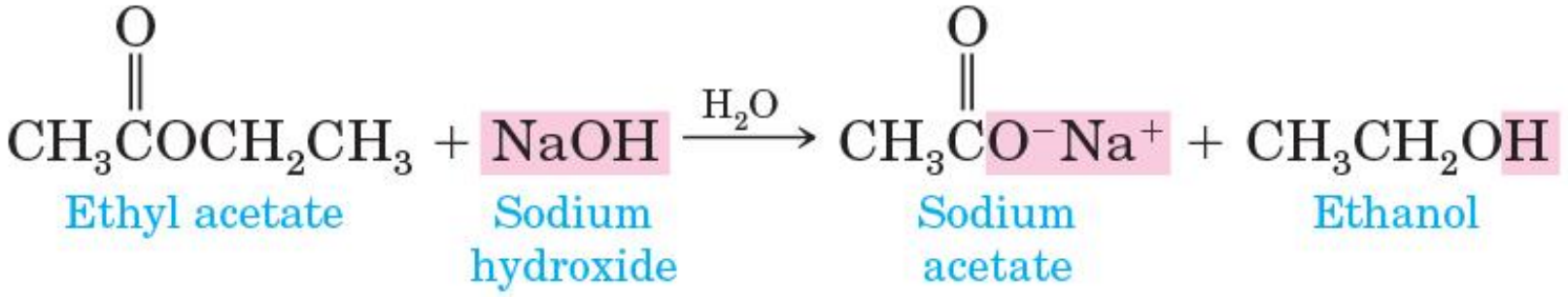
- Carboxylic anhydrides, particularly the low-molecular- weight ones, react readily with water (hydrolyze) to give two carboxylic acids.



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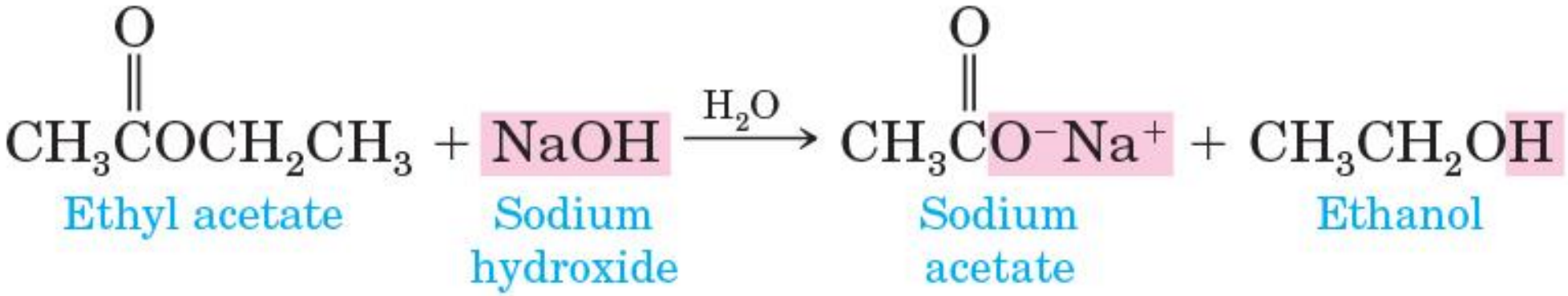
Hydrolysis of Esters

- Esters hydrolyze only very slowly, even in boiling water.
- Hydrolysis becomes considerably more rapid, however, when the ester is heated in aqueous acid or base.
- Hydrolysis of esters in aqueous acid is the reverse of Fischer esterification.
- A large excess of water drives the equilibrium to the right to form the carboxylic acid and alcohol (Le Chatelier's principle).



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- We can also hydrolyze an ester using a hot aqueous base, such as aqueous NaOH.
- This reaction is often called **saponification**, a reference to its use in the manufacture of soaps.
- The carboxylic acid formed in the hydrolysis reacts with hydroxide ion to form a carboxylic acid anion.
- Each mole of ester hydrolyzed requires one mole of base.

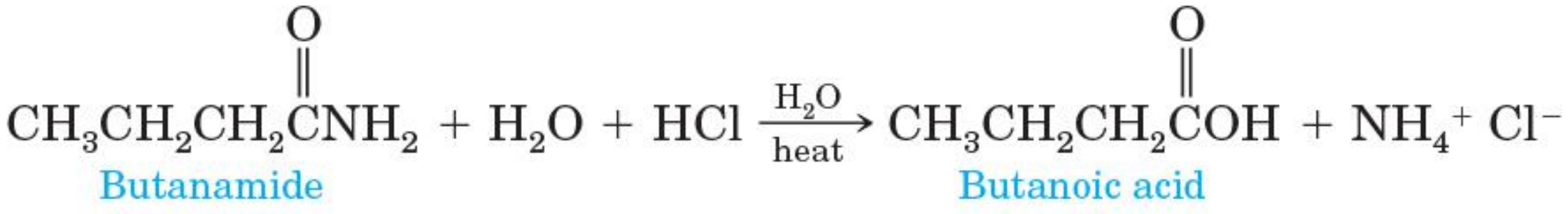


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Hydrolysis of Amides

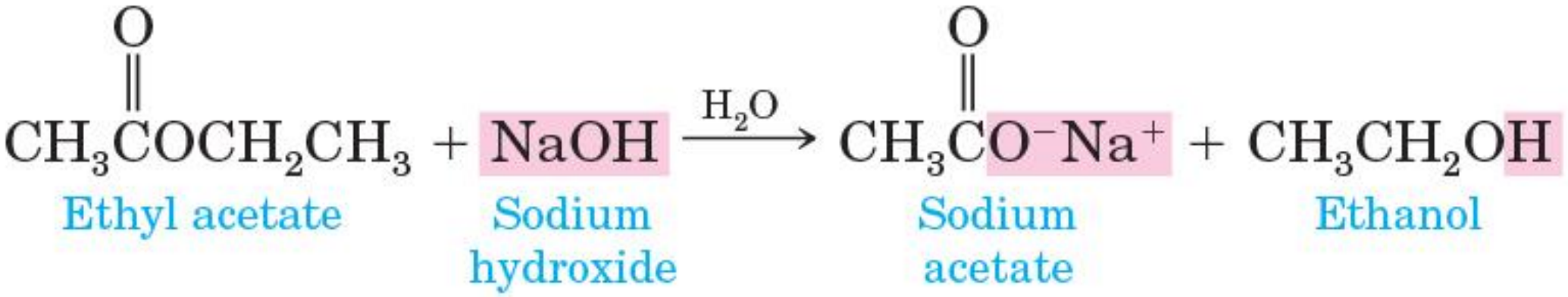
Amides require more vigorous conditions for hydrolysis in both acid and base than do esters.

- Hydrolysis in hot aqueous acid gives a carboxylic acid and an ammonium ion.
- Hydrolysis is driven to completion by the acid–base reaction between ammonia or the amine and the acid to form an ammonium ion.
- Each mole of amide hydrolyzed requires one mole of acid.



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- Hydrolysis of an amide in aqueous base gives a carboxylic acid salt and ammonia or an amine.
- Hydrolysis is driven to completion by the acid-base reaction between the carboxylic acid and base to form a salt.
- Each mole of amide hydrolyzed requires one mole of base.

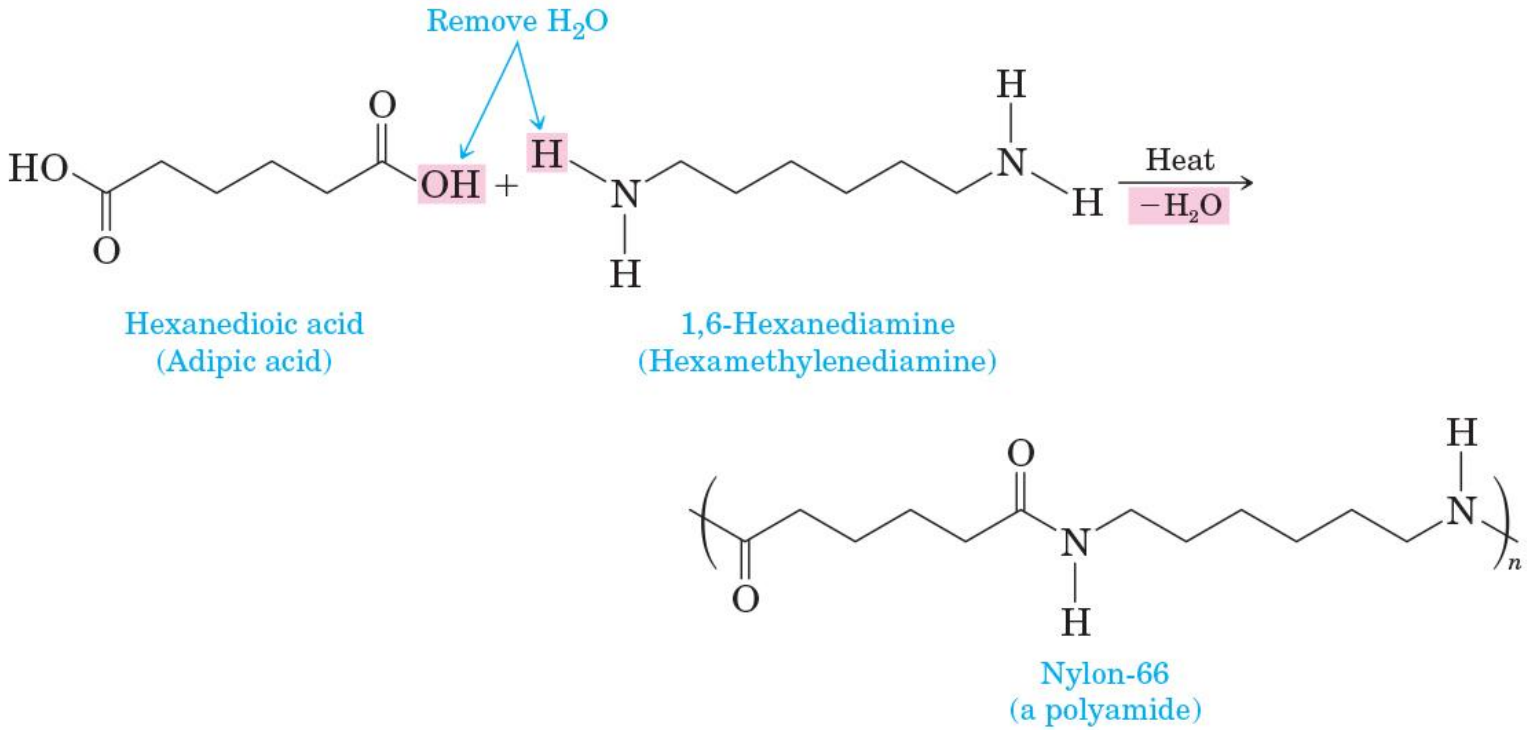


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Polyamides

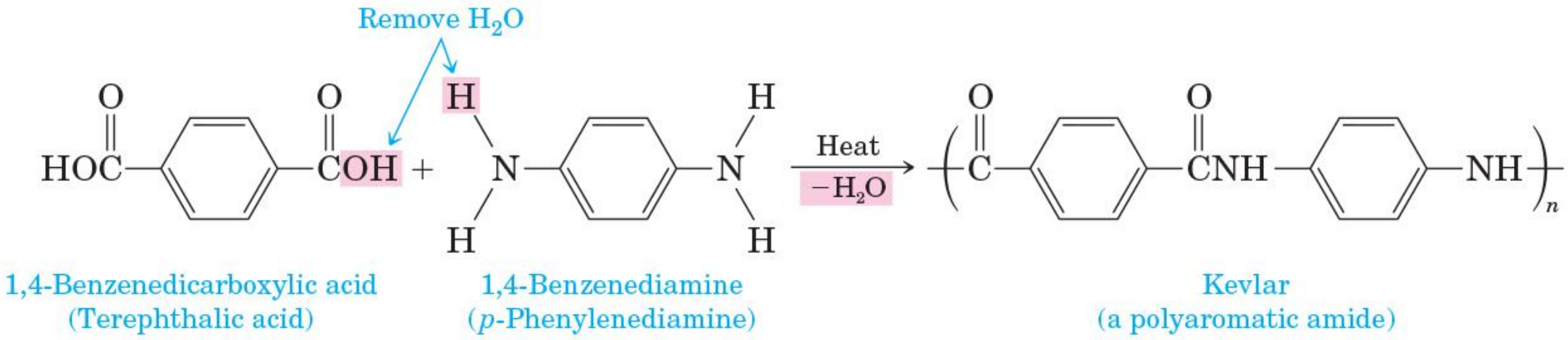
Nylon-66 was the first purely synthetic fiber.

- It is synthesized from two six-carbon monomers.



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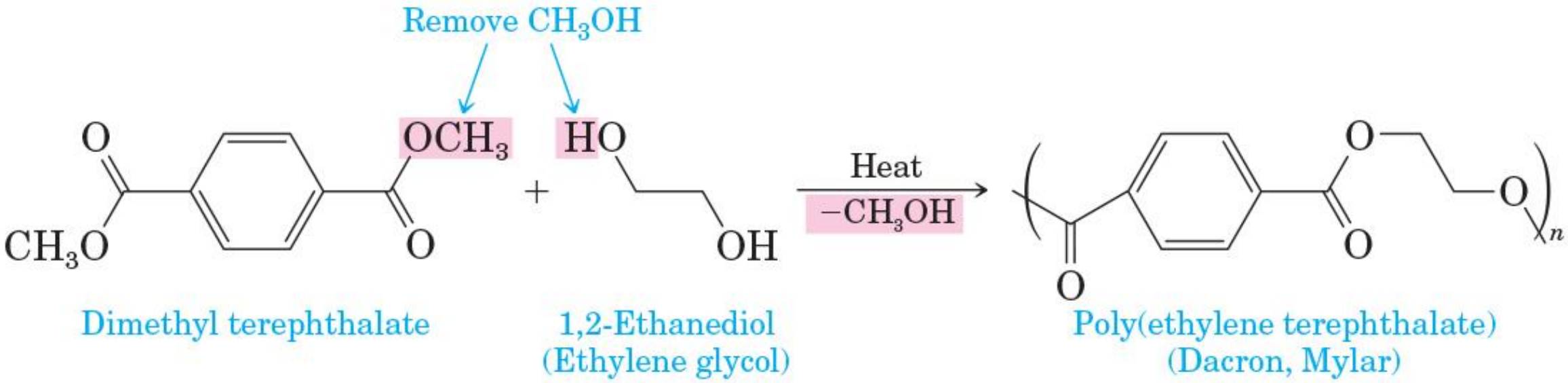
The polyaromatic amide known as **Kevlar** is made from an aromatic dicarboxylic acid and an aromatic diamine.



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Polyesters

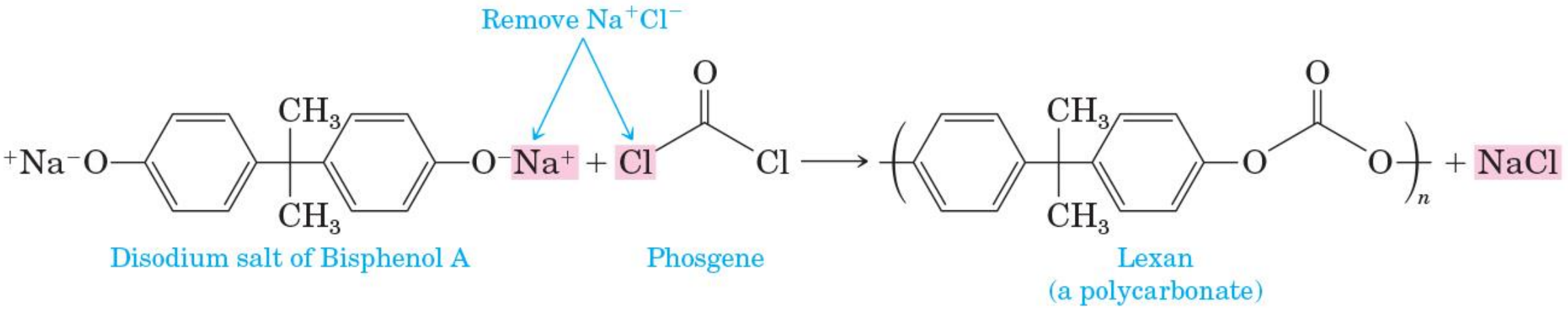
The first synthetic polyester involved polymerization of this diester and diol.



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Polycarbonates

- Lexan, the most familiar **polycarbonate**, is formed by reaction between the disodium salt of bisphenol A (BPA) and phosgene.



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